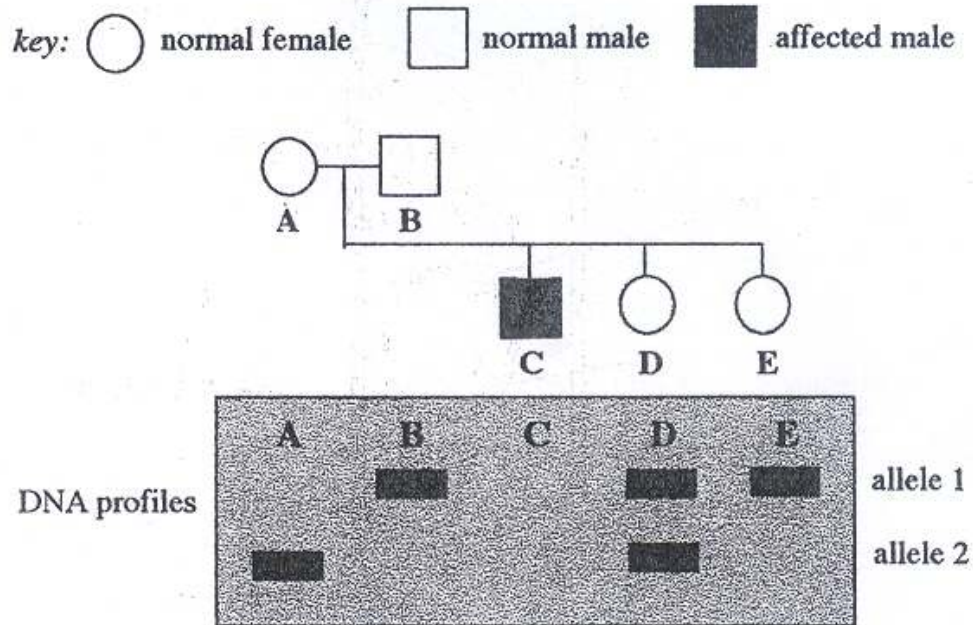


Section AAnswer **ALL** questions in this section.**Question 1**

DNA analysis may be used in genetic engineering. A large deletion in a gene can be detected by the absence of specific DNA bands in a DNA profile when a probe for that portion of the gene has been used. DNA from a boy suffering from a sex-linked genetic disease caused by a large deletion was analysed, as was that of his parents and two sisters. The family's pedigree and DNA profiles are shown in Fig. 1.1.

**Fig. 1.1**

(a) Describe briefly how, in such DNA analysis,

(i) DNA is separated into bands;

[3]

(ii) the bands are made visible.

[2]

(b) Explain why the profiles of

(i) the boy's mother (**A**) and father (**B**) each show only one allele;

[2]

(ii) the boy (**C**) shows neither allele.

[1]

Haemophilia A, an X-linked disorder of blood clotting, is caused by a mutation in glycoprotein clotting factor called Factor VIII. Many single nucleotide mutations of this gene have been described, making the detection of mutant genes by Southern blots inefficient. There is however, an RFLP for the enzyme *Hind*III contained in an intron of the Factor VIII gene that can often be used in screening, as shown in the Fig. 1.2 below.

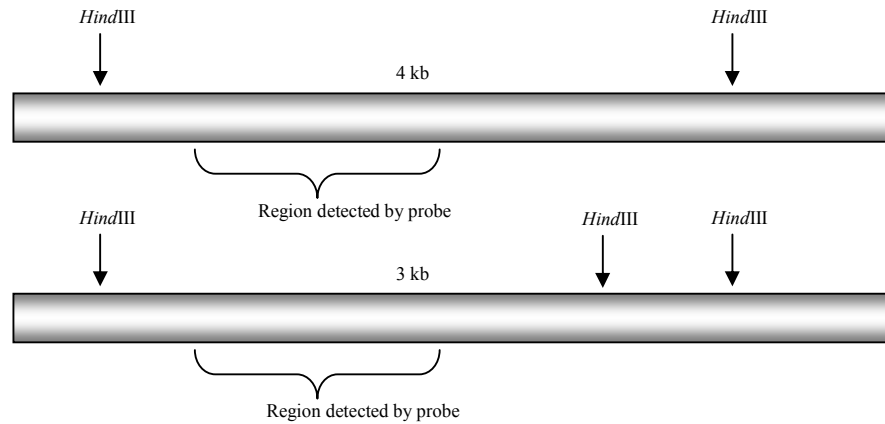


Fig. 1.2

DNA is obtained from a family who has a son with haemophilia and is cut with *Hind*III and the fragments are probed and visualised by Southern Blotting. The results are shown in Fig. 1.3.

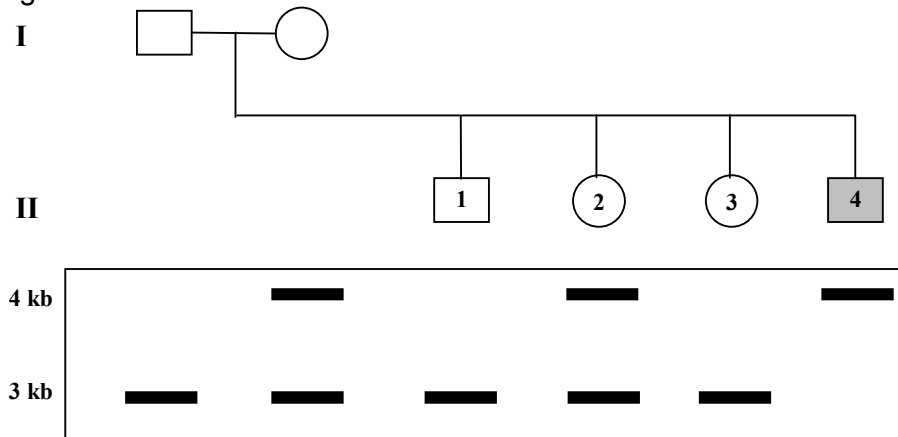


Fig. 1.3

(c) With reference to Fig. 1.3,

(i) state and explain which of the female(s) is/ are carriers of haemophilia. [3]

(ii) explain why there is a difference in the number of bands produced from individual 1 – 4. [4]

(d) Explain why single nucleotide mutations of the gene make detection of mutant genes by Southern Blotting inefficient. [2]

[Total: 17]

Question 2

The first British trial of a possible gene therapy for cystic fibrosis (CF) began in 1993. Nine adult male patients with advanced lung disease were chosen. Six patients received gene therapy and three acted as a control.

The normal allele of the CF gene codes for a protein that forms an ion channel in the membrane of the epithelial cells that line the gut and the air-ways. In sufferers of CF, chloride ions cannot pass through this channel.

The purpose of the trial was to see whether functioning ion channels could be manufactured by cells which have had the normal allele of the CF gene inserted into them. DNA coding for the normal allele was cloned and put into small spheres of phospholipids bilayers (liposomes). The droplets were inhaled as an aerosol into the nose. Liposomes were thought to be a safer vector than the adenoviruses used in earlier trials in North America.

Gene therapy cannot provide a permanent cure for cystic fibrosis since cells from the airways are shed constantly.

Male patients were chosen for the British trial to avoid the risks of the allele being passed into the germ-line, since virtually all men with CF are infertile.

- (a) Explain how cystic fibrosis is inherited. [3]

- (b)(i) Suggest how a liposome can be a suitable vector for introducing DNA into mammalian cells. [2]

-
-
-
-
-

[illegible]

(c) With reference to Fig. 2.1,

-

- (ii) state the effect this difference would produce in the polypeptide coded for by the gene. [1]

- (iii) Explain how individual **P** could also suffer from CF. [2]

Heterozygotes for the most common mutant allele causing CF may be more resistant to bacterial infections of the gut than homozygotes for the normal allele.

- (d) Explain how such resistance could affect the frequency of the mutant allele in a population. [2]

[Total: 14]

Question 3

By 1999, genetically modified maize plants replaced non-genetically modified maize plants as the more widely grown crop in the USA. One of the transgenic varieties of maize contains a gene (*Bt*) from a bacterium, *Bacillus thuringensis*. The gene codes for a toxin, which is expressed in the leaves and acts as an insecticide.

- (a)(i) Explain what is meant by a *transgenic variety of maize*. [2]

- (ii) why farmers in the USA grow maize carrying the *Bt* gene. [2]

- (b) Maize is a monocotyledonous plant which *Agrobacterium* would not infect. Suggest one method by which the *Bt* gene can be introduced into the plant. [1]

The *Bt* gene is also expressed in pollen, which may be dispersed by wind, at least 60m away from the parent plant. In the USA, milkweed frequently grows around the edge of maize fields and is fed upon by caterpillars of the Monarch butterfly.

In an experiment conducted to investigate the environmental effects of *Bt* maize, leaves of the milkweed were divided into three groups, **A**, **B** and **C**, and treated in the following ways:

- A** : dusted with pollen from genetically modified maize carrying the *Bt* gene;
- B** : dusted with pollen from maize that had not been genetically modified;
- C** : not dusted with pollen.

Monarch caterpillars were then placed on the leaves and the mean area of leaf eaten per caterpillar and the survival of the caterpillars was measured over four days. The results of the investigation are shown in Fig. 3.1 and 3.2.

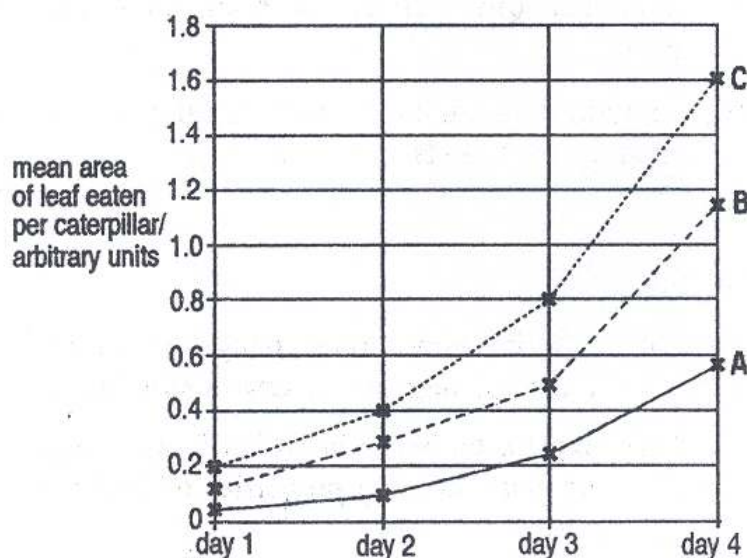


Fig. 3.1

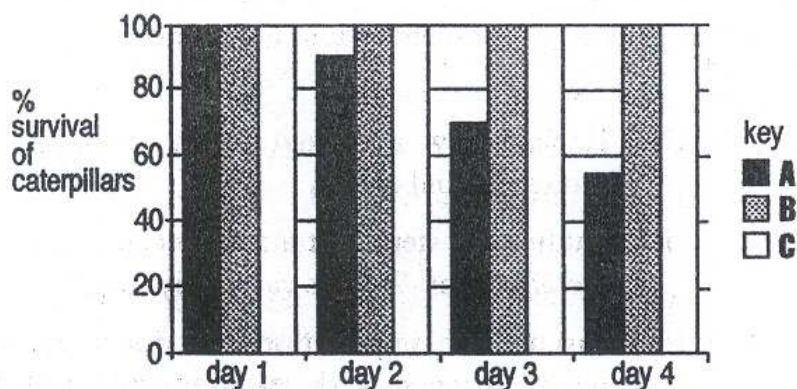


Fig. 3.2

(c) With reference to Figs. 3.1 and 3.2,

- (i) comment on the effect of consuming pollen from genetically modified maize carrying the *Bt* gene on the mean area of leaf eaten per caterpillar; [4]

- (ii) compare the effect of consuming pollen from different maize plants on the survival of Monarch butterfly caterpillars. [2]

In the USA, 50% of the summer population of Monarch butterflies is found in the maize-growing area. Maize pollen is shed between late June and the middle of August, which is during the period when Monarch caterpillars are feeding.

- (d) Using only information given in this question, predict with reasons, the effect of continued cultivation of *Bt* maize in the USA on the population of Monarch butterflies. [3]

[Total : 14]

Section B

Answer **ALL** questions.

Write your answers on the separate writing papers provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 4

- (a)** Describe how bacteria can be genetically engineered to produce a named human protein.

[10]

- (b)** Using appropriate examples, explain the normal functions of stem cells in living organisms.

[6]

- (c)** Outline the ethical and social implications of genetically modified organisms.

[4]

[Total: 20]

End of H2 Paper 3